Week 3

Wednesday, February 04, 2026 9:12 PM

Module 8 – Properties of Matrix Multiplication

Theorem 1. *Let A be an $m \times n$ matrix and let B and C be matrices for which the indicated sums and products are defined.*

(a) $A(BC) = (AB)C$ (associative law of multiplication)
(b) $A(B+C) = AB + AC$, $(B+C)A = BA + CA$ (distributive laws)
(d) $r(AB) = (rA)B = A(rB)$ for every scalar r,
(e) $A(rB + sC) = rAB + sAC$ for every scalars r, s (linearity of matrix multiplication)
(f) $I_m A = A = AI_n$ (identity for matrix multiplication)

For the definition of Ak for A.... A k times that is Ak is obtained by multiplying A k-times with itself, it would only make sense of A is any m x n matrix but m = n.

Module 9 – Elementary matrices

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix.

A permutation matrix is one that is obtained by performing row exchanges on an identity matrix.

A **permutation matrix** is one that is obtained by performing row exchanges on an identity matrix.

Example. Consider $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$. Are these matrices elementary matrices?

Solution.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \underset{R_2 \to 3R_2}{\rightsquigarrow} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \underset{R_2 \leftrightarrow R_3}{\rightsquigarrow} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The matrices on the left of the operation are identity matrix

Theorem 1. *If an elementary row operation is performed on an $m \times n$-matrix A, the resulting matrix can be written as EA, where the $m \times m$-matrix E is created by performing the same row operations on I_m.*

Basically, what this means that elementary row operation performed on a specific matrix is equivalent to if I were to multiply that matrix by a row = column matrix that is created through having that same row operation.

I_m should be Identity matrix m x m.

Something like this:

I_n is the $n \times n$-matrix

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Theorem 2. *Let A, B be two $m \times n$-matrices and row-equivalent. Then there is a sequence $m \times m$-elementary matrices $E_1, \ldots, E_\ell$ such that*

$$E_\ell \ldots E_1 A = B.$$

Example. Consider $A = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Find two elementary matrices E_1, E_2 such that $E_2 E_1 A = B$.

Solution.

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 4 \end{bmatrix} \underset{R_1 \leftrightarrow R_2}{\rightsquigarrow} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 4 \end{bmatrix} \underset{R_3 \leftrightarrow R_3 - 2R_1}{\rightsquigarrow} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = B$$

$$\text{Set } E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

$$E_2 E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Module 10 – Inverse of a Matrix

Not all real numbers have inverse. 0 to the negative on is not well defined.

We say that a matrix A is said to be invertible If there is an n x n matrix C satisfying
CA = AC = In

Elementary matrices are invertible, because row operations are reversible.

Theorem 2. *Suppose A and B are invertible. Then:*
(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$ (i.e. A is the inverse of A^{-1}).
(b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
(c) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Question. If A is an invertible $n \times n$ matrix, then A has how many pivot columns? [A|b]

A must have n pivots, because otherwise $A\mathbf{x} = \mathbf{b}$ would not have a solution of each $\mathbf{b}$.

If there's n-1 pivot, that mean there will be a row where it looks like this [0 0 0 0 0 | b]

Module 11 – Computing the inverse

Theorem 1. *Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

Not shown, but place an identity matrix next to the matrix you want to find the inverse of. Find the RREF of the matrix, and apply whatever row operations you used to the identity matrix. There, you got your inverse.